Fiscal Year:	FY 2015	**Task Last Updated:**	FY 05/26/2015
PI Name:	Bloom, David Ph.D.		
Project Title:	Effect of Spaceflight on Herpesvirus Genome Stability and Diversity		
Division Name:	Human Research, Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	Gainesville	**State:**	FL
Zip Code:	32610-3003	**Congressional District:**	6
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight (ILSRA) NNH14ZTT002N
Start Date:	06/01/2015	**End Date:**	06/30/2016
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
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Flight Program:			
Flight Assignment:	NOTE: End date is 6/30/2016 per NSSC information (Ed., 12/6/16) NOTE: Grant ended early, per F. Hernandez/ARC (Ed., 5/4/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	McIntyre, Lauren Ph.D. (University of Florida) Mehta, Satish Ph.D. (Wyle Laboratories, Inc.) Pierson, Duane Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	NNX15AL04G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE this project is jointly funded by NASA's Space Biology and Human Research Program for collaborative purposes. Viral reactivation, especially of the Herpesviruses, is known to be increase during spaceflight. At least four (EBV, VZV, CMV, and HSV1) of the eight known human Herpesviruses are common life-long infections of humans have been shown to reactivated due to spaceflight. The fact that spaceflight is known to increase the frequency and duration of Herpesvirus reactivation sets up a biological environment for these reactivating and replicating viruses to mutate. Recently it has been shown that environmental factors such as diet and medications can alter the mutation rate and pathogenesis of human viruses, so it seems likely that effects of microgravity, radiation, and other physiological aspects of spaceflight may have a profound effect of on the evolution of mutations and adaptations of the Herpesviruses during spaceflight. Therefore the goal of the proposed team study will be to determine the changes in the genomic and mutational diversity that is present in the Herpesvirus virome present in astronaut saliva and urine samples collected before, during, and after spaceflight. In order to determine the effect of spaceflight on the generation of mutations and diversity of the Herpesviruses we will: 1) Collect and analyze astronaut saliva and urine samples for Herpesvirus shedding, 2) Perform deep-sequence analysis of the Herpesviruses present in the saliva of astronauts before, during and after spaceflight, and 3) Analyze these data to determine the overall diversity of the viruses present in the samples before, during and after spaceflight and the mutation frequency present in the samples before, during, and after spaceflight. In summary, the proposed study is a unique opportunity to examine the influence of spaceflight on the mutation rate and diversity of human persistent viruses and to assess the potential risk of virus mutations accumulating during long-duration missions. The overall PI/PD (Principal Investigator/Project Director) of this team project will be Dr. David Bloom at the University of Florida who will also direct Project 2 which includes the library construction and sequencing of the viral DNAs isolated in saliva. Validated astronaut saliva samples will be provided by Drs. Satish Mehta and Duane Pierson of NASA Johnson Space Center (JSC). As part of Project 1, Drs. Mehta and Pierson of JSC will also apply enhanced recovery and PCR (polymerase chain reaction) technology to more sensitively measure herpesvirus shedding during spaceflight. Dr. Lauren McIntyre will serve as a Co-I at University of Florida (UF) for Project 2 in supervising the bioinformatics analyses of the sequenced herpesviruses for mutation rates. This study is significant as it could identify the potential for risks for these viruses generating mutations that could enhance virulence or reactivation propensity that would pose a risk to astronauts' health. Such a risk could be significant especially for spaceflights of longer durations. In addition to providing this key insight as a means of understanding the risks to astronauts’ health, it could provide important new data for terrestrial use related to the mutation rates of these persistent human viruses when exposed to stress stimuli that may accelerate mutation rates.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In addition to providing this key insight as a means of understanding the risks to astronauts’ health, the research could provide important new data for terrestrial use related to the mutation rates of these persistent human viruses when exposed to stress stimuli that may accelerate mutation rates.
Task Progress:	New project for FY2015.
Bibliography Type:	Description: (Last Updated:)